

Animal Kingdom

(Protozoa to Mammalia)

60 HARD-level MCQs covering all 19 topics of the syllabus

With answer key & solutions

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Basis of Classification

Question 1

HARD

In enterocoelous coelom formation, the coelom is derived by:

- A. Splitting of solid mesodermal blocks (schizocoely)
- B. Outpouching of the archenteron wall that pinches off as coelomic pouches ✓**
- C. Persistence of the embryonic blastocoel
- D. Hollowing out of a solid mass of mesodermal cells

Solution: In enterocoelous development (seen in Echinodermata and Chordata), the coelom arises as pouches budding off the archenteron (embryonic gut) wall, unlike schizocoely where mesodermal blocks split to form the cavity.

Question 2

HARD

The key criterion that distinguishes the pseudocoelomate condition (e.g., Aschelminthes) from the true coelomate condition (e.g., Annelida) is that in pseudocoelomates:

- A. The body cavity is completely lined by mesoderm on both the body-wall and gut sides
- B. The body cavity is present between the mesoderm and endoderm, lined by mesoderm only on the body-wall side ✓**
- C. There is no body cavity between the body wall and gut
- D. The body cavity is filled entirely with parenchyma

Solution: In pseudocoelomates, the body cavity is not a true coelom; it is lined by mesoderm only on the body-wall side and lacks mesodermal lining around the gut, unlike the fully mesoderm-lined true coelom of annelids.

Question 3

HARD

Match Column-I (Phylum) with Column-II (Type of body cavity) and select the correct option.

Column-I: (a) Platyhelminthes (b) Aschelminthes (c) Annelida (d) Arthropoda

Column-II: (i) Acoelomate (ii) Pseudocoelomate (iii) True coelom (schizocoelous) (iv) Haemocoel (reduced true coelom)

A. a-i, b-ii, c-iii, d-iv ✓

B. a-ii, b-i, c-iv, d-iii

C. a-i, b-iii, c-ii, d-iv

D. a-iii, b-ii, c-i, d-iv

Solution: Platyhelminthes = acoelomate; Aschelminthes = pseudocoelomate; Annelida = true coelomate (schizocoelous); Arthropoda = haemocoel, a reduced coelom filled with blood, associated with open circulation.

Phylum Protozoa

Question 4

HARD

Which of the following statements about Sporozoans (e.g., Plasmodium) is INCORRECT?

A. They completely lack locomotory organelles.

B. Infective spore-like stages are found in their life cycle.

C. They possess prominent contractile vacuoles used for locomotion. ✓

D. Plasmodium, the malarial parasite, is a well-known sporozoan.

Solution: Contractile vacuoles (where present in protozoans) function in osmoregulation, not locomotion; Sporozoans are additionally characterised by a complete absence of locomotory organelles at the infective stage.

Question 5

HARD

Assertion (A): Plasmodium causes malaria in humans.

Reason (R): It undergoes sexual reproduction inside the human host and asexual multiplication inside the female Anopheles mosquito.

- A. Both A and R are true and R is the correct explanation of A.
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false. ✓
- D. A is false but R is true.

Solution: The Reason is reversed. Plasmodium undergoes asexual multiplication (schizogony) in humans and sexual reproduction (sporogony) in the female Anopheles mosquito, not the other way round.

Question 6

HARD

Which of the following correctly matches the protozoan class with its characteristic locomotory structure?

- A. Amoeboid protozoans – Cilia
- B. Flagellated protozoans – Pseudopodia
- C. Ciliated protozoans – Flagella
- D. Ciliated protozoans – Cilia ✓

Solution: Amoeboid protozoans move by pseudopodia, flagellated protozoans by flagella, and ciliated protozoans (e.g., Paramecium) by cilia — so the only correctly matched pair is Ciliated protozoans – Cilia.

Phylum Porifera

Question 7

HARD

Which canal system, characteristic of the MAJORITY of sponges, provides the greatest surface area for water filtration through extensive folding of the body wall?

A. Ascon type

B. Sycon type

C. Leucon (Rhagon) type ✓

D. None of the sponges show a folded canal system

Solution: The leuconoid (Rhagon) canal system, with numerous flagellated chambers, is the most complex and provides the largest surface area for filtration; it is found in most sponges.

Question 8

HARD

Assertion (A): Sponges can survive unfavourable conditions through structures called gemmules.

Reason (R): Gemmules consist of an amoebocyte mass surrounded by a protective, spicule/spongin-reinforced covering, formed characteristically in freshwater sponges.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct — gemmules are internal asexual buds with a protective covering that allow sponges (notably freshwater forms like *Spongilla*) to survive harsh conditions, which correctly explains the assertion.

Question 9

HARD

Which of the following genera possesses an endoskeleton made ONLY of spongin fibres, without any mineral spicules?

A. *Spongilla*

B. *Sycon*

C. *Euspongia* ✓

D. *Hyalonema*

Solution: *Euspongia* (bath sponge) has a skeleton composed purely of spongin fibres. *Spongilla* has siliceous spicules, *Sycon* has calcareous spicules, and *Hyalonema* has siliceous spicules (glass rope sponge).

Coelenterata & Ctenophora

Question 10

HARD

Assertion (A): Obelia exhibits alternation of generation (metagenesis).

Reason (R): Its life cycle alternates between an asexual polyp stage and a sexual medusa stage.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct, and the alternation between the asexually-reproducing polyp and sexually-reproducing medusa forms is precisely what defines metagenesis in Obelia.

Question 11

HARD

Which of the following statements about phylum Ctenophora is INCORRECT?

A. Locomotion occurs by means of eight rows of ciliary comb plates.

B. They exhibit biradial symmetry.

C. Digestion in ctenophores is exclusively intracellular. ✓

D. All ctenophores are exclusively marine and often bioluminescent.

Solution: Ctenophores show BOTH extracellular (in the gastrovascular cavity) and intracellular digestion — digestion is not exclusively intracellular.

Question 12

HARD

Which cnidarian class is correctly matched with an example that exists ONLY in the polyp form, with no medusa stage in its life cycle?

A. Hydrozoa – Physalia

B. Scyphozoa – Aurelia

C. Anthozoa – Adamsia ✓

D. Hydrozoa – Aurelia

Solution: Class Anthozoa (e.g., Adamsia the sea anemone, and corals) exists exclusively as polyps; the medusa stage is entirely absent, unlike Hydrozoa and Scyphozoa.

Phylum Platyhelminthes

Question 13

HARD

Assertion (A): *Taenia solium* (tapeworm) lacks a digestive system altogether.

Reason (R): It absorbs already-digested food materials from the host's intestine directly through its body surface.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct — as an intestinal endoparasite bathed in digested host food, *Taenia* has no need for its own digestive system and absorbs nutrients across its body surface.

Question 14

HARD

Flame cells, a characteristic structure of phylum Platyhelminthes, primarily function in:

A. Respiration

B. Osmoregulation and excretion ✓

C. Locomotion

D. Sexual reproduction

Solution: Flame cells are specialised excretory/osmoregulatory cells that help remove excess water and nitrogenous wastes, maintaining internal fluid balance.

Question 15

HARD

Which of the following statements about *Fasciola hepatica* (liver fluke) is INCORRECT?

A. It is an endoparasite infecting the liver of sheep and other herbivores.

B. Its life cycle involves a snail as an intermediate host.

C. It possesses a well-developed true coelom. ✓

D. It shows a flattened, leaf-like body typical of trematodes.

Solution: Like all Platyhelminthes, *Fasciola* is acoelomate — it lacks a true coelom; the space between body wall and internal organs is filled with parenchyma.

Phylum Aschelminthes

Question 16

HARD

Which of the following is a correct feature of phylum Aschelminthes?

- A. Organisms are typically hermaphroditic.
- B. The body cavity is a true, mesoderm-lined coelom.
- C. Sexes are usually separate (dioecious), often with distinct sexual dimorphism. ✓**
- D. The body is metamerically segmented.

Solution: Aschelminthes (e.g., Ascaris) are typically dioecious with sexual dimorphism (males and females often differ in size/tail shape); the body cavity is a pseudocoelom, and there is no true metameric segmentation.

Question 17

HARD

Assertion (A): *Wuchereria bancrofti* causes filariasis, marked by chronic inflammation of the lymphatic vessels.

Reason (R): The parasite is transmitted to humans through the bite of an infected female *Culex* mosquito.

- A. Both A and R are true and R is the correct explanation of A. ✓**
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false.
- D. A is false but R is true.

Solution: Both statements are correct; *Culex* mosquitoes are the vectors that transmit the filarial worm to humans, directly accounting for the disease's mode of spread.

Question 18

HARD

Ascaris lumbricoides differs from *Fasciola hepatica* most fundamentally in possessing:

A. A pseudocoelom, in contrast to the acoelomate condition of *Fasciola* ✓

B. Flame cells for excretion

C. A hermaphroditic reproductive system

D. Bilateral symmetry

Solution: The key distinguishing feature between the two phyla is body cavity type: Aschelminthes (*Ascaris*) is pseudocoelomate, while Platyhelminthes (*Fasciola*) is acoelomate. Both groups actually share bilateral symmetry, so that is not a distinguishing feature.

Phylum Annelida

Question 19

HARD

Which of the following correctly distinguishes *Nereis* from *Hirudinaria* (leech)?

A. *Nereis* possesses parapodia for locomotion; *Hirudinaria* lacks parapodia. ✓

B. *Nereis* is a freshwater form; *Hirudinaria* is exclusively marine.

C. *Nereis* is monoecious; *Hirudinaria* is dioecious.

D. *Nereis* lacks a closed circulatory system; *Hirudinaria* possesses one.

Solution: *Nereis* (a marine polychaete) has lateral parapodia used in locomotion, while *Hirudinaria* (leech, a freshwater/terrestrial annelid) lacks parapodia and instead uses suckers for movement.

Question 20

HARD

Assertion (A): Annelids possess a closed type of circulatory system.

Reason (R): Blood flows through well-defined vessels and does not directly bathe the body tissues.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the defining feature of a closed circulatory system is that blood remains confined to vessels rather than directly bathing tissues (as it does in an open system).

Question 21

HARD

Nephridia, the characteristic segmentally-arranged excretory organs of Annelida, primarily function in:

A. Osmoregulation and excretion of nitrogenous wastes ✓

B. Digestion of ingested food

C. Gaseous exchange with the environment

D. Sexual reproduction

Solution: Nephridia are tubular structures that help maintain osmotic balance and eliminate nitrogenous excretory wastes from the coelomic fluid.

Phylum Arthropoda

Question 22

HARD

Which of the following respiratory structures is the hallmark of class Arachnida (e.g., spiders and scorpions)?

A. Book gills

B. Book lungs ✓

C. Tracheae exclusively

D. Ctenidia (gills)

Solution: Book lungs, thin membranous folds housed in a chamber, are the characteristic respiratory organs of most arachnids, distinguishing them from the tracheal system typical of insects.

Question 23

HARD

Assertion (A): Arthropoda is the largest phylum in the animal kingdom.

Reason (R): Over two-thirds of all named animal species on Earth belong to Arthropoda.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the sheer number and diversity of named arthropod species is precisely the evidence for, and explanation of, its status as the largest animal phylum.

Question 24

HARD

Peripatus is considered an important 'missing link' connecting which two phyla, based on its mix of characters?

A. Annelida and Arthropoda ✓

B. Mollusca and Arthropoda

C. Annelida and Mollusca

D. Arthropoda and Chordata

Solution: Peripatus shows annelid-like features (thin unjointed cuticle, unsegmented internal organisation) combined with arthropod-like features (jointed appendages), making it an evolutionary link between Annelida and Arthropoda.

Question 25

HARD

Which of the following correctly describes the circulatory system of Arthropoda?

A. Closed type, with blood always confined to definite vessels

B. Open type, in which the body cavity is termed a haemocoel ✓

C. Arthropods entirely lack a circulatory system

D. A single vessel circulates blood strictly unidirectionally with no body cavity involved

Solution: Arthropods have an open circulatory system — the haemolymph (blood) is pumped into the haemocoel (a blood-filled body cavity) where it directly bathes the tissues, rather than remaining confined to vessels.

Question 26

HARD

Match the arthropod class with its typical respiratory organ and select the correct option.

Column-I: (a) Insecta (b) Crustacea (c) Arachnida

Column-II: (i) Book lungs (ii) Tracheae (iii) Gills

A. a-ii, b-iii, c-i ✓

B. a-i, b-ii, c-iii

C. a-iii, b-i, c-ii

D. a-ii, b-i, c-iii

Solution: Insects respire through tracheae, most crustaceans through gills, and arachnids typically through book lungs → a-ii, b-iii, c-i.

Phylum Mollusca

Question 27

HARD

Which of the following molluscan classes characteristically LACKS a radula?

A. Gastropoda

B. Cephalopoda

C. Pelecypoda (Bivalvia) ✓

D. Polyplacophora

Solution: Bivalves (Pelecypoda), being filter feeders, lack a radula — the rasping, tongue-like feeding organ found in most other molluscan classes.

Question 28

HARD

Assertion (A): Cephalopods such as Octopus and Sepia show highly developed nervous systems and complex behaviour.

Reason (R): In cephalopods, the foot is modified into tentacles surrounding the head region.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is NOT the correct explanation of A. ✓

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are factually correct, but the modification of the foot into tentacles does not causally explain the advanced development of the cephalopod nervous system — so R is not the correct explanation of A.

Question 29

HARD

The mantle cavity in molluscs typically houses which pair of structures?

A. Ctenidia (gills) along with the openings of the reproductive, excretory and digestive systems ✓

B. Radula and foot only

C. Heart and nephridia only

D. Brain ganglia and eyes only

Solution: The mantle cavity is a key molluscan feature that houses the ctenidia (used in respiration) as well as the outlets of the excretory, digestive and reproductive systems.

Phylum Echinodermata

Question 30

HARD

The unique water vascular system of Echinodermata primarily functions in:

A. Locomotion ✓

B. Reproduction

C. Excretion

D. Digestion

Solution: The water vascular system, operating through tube feet, is the principal means of locomotion in echinoderms (and also assists in food capture in some forms).

Question 31

HARD

Assertion (A): Echinoderms are triploblastic, coelomate animals with organ-system level of organisation.
Reason (R): Their coelom develops through the enterocoelous method, a feature shared with Chordata and considered evolutionarily significant.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the shared enterocoelous mode of coelom formation between Echinodermata and Chordata is a key piece of evidence for their evolutionary proximity.

Question 32

HARD

Which of the following is a sessile, stalked echinoderm commonly known as the 'sea lily'?

A. Asterias

B. Antedon ✓

C. Echinus

D. Holothuria

Solution: Antedon (sea lily/feather star) is a stalked, often sessile echinoderm, unlike the free-living Asterias (starfish), Echinus (sea urchin) and Holothuria (sea cucumber).

Phylum Hemichordata

Question 33

HARD

Balanoglossus was historically placed within phylum Chordata because of the presence of a structure called the:

- A. Notochord
- B. Stomochord (buccal diverticulum) ✓**
- C. Post-anal tail
- D. Dorsal tubular nerve cord

Solution: Balanoglossus possesses a structure in its proboscis called the stomochord, once mistaken for a notochord; it is now known not to be homologous, so Hemichordata is treated as a separate phylum.

Question 34

HARD

Assertion (A): Hemichordata is now considered a separate phylum, distinct from Chordata.

Reason (R): The stomochord present in the proboscis of Balanoglossus is not a true notochord.

- A. Both A and R are true and R is the correct explanation of A. ✓**
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false.
- D. A is false but R is true.

Solution: Both statements are correct; the discovery that the stomochord is not homologous to a true notochord is exactly why Hemichordata was reclassified out of Chordata.

Chordata – General Characters

Question 35

HARD

Which of the following is NOT one of the four fundamental diagnostic characters shared by all chordates at some stage of life?

A. Notochord

B. Dorsal tubular nerve cord

C. Ventrally-placed heart ✓

D. Paired pharyngeal gill slits

Solution: The four classical diagnostic chordate characters are: notochord, dorsal tubular nerve cord, paired pharyngeal gill slits, and a post-anal tail. A ventral heart, while typical of chordates, is not one of these four defining features.

Question 36

HARD

Assertion (A): All chordates are bilaterally symmetrical, triploblastic and coelomate.

Reason (R): These same three features are also shared by several non-chordate phyla such as Annelida and Mollusca.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is NOT the correct explanation of A. ✓

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are true independently, but R (that non-chordates also share these features) does not explain why chordates themselves possess them — so R is not the correct explanation of A.

Protochordata

Question 37

HARD

In which subphylum of Protochordata does the notochord persist throughout life, extending from the head to the tail region?

A. Urochordata

B. Cephalochordata ✓

C. Vertebrata

D. Hemichordata

Solution: In Cephalochordata (e.g., Branchiostoma/Amphioxus), the notochord extends from head to tail and persists throughout life, unlike Urochordata where it is present only in the larval tail.

Question 38

HARD

Assertion (A): Protochordates are exclusively marine animals.

Reason (R): In Urochordata, the notochord is present only in the larval tail and is lost in the adult.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is NOT the correct explanation of A. ✓

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are true, but the loss of the larval notochord in adult urochordates has no bearing on why protochordates are marine — the two facts are unrelated, so R does not explain A.

Vertebrata – Cyclostomata

Question 39

HARD

Which of the following statements about Cyclostomata is INCORRECT?

A. They possess a cartilaginous cranium and vertebral column.

B. They are ectoparasites on certain fishes.

C. They possess well-developed jaws for biting into host tissue. ✓

D. Petromyzon (lamprey) is a well-known example.

Solution: Cyclostomes are jawless vertebrates (Agnatha) — they possess a circular, sucker-like mouth for attaching to and rasping host tissue, not true jaws.

Question 40

HARD

Marine lampreys such as *Petromyzon* migrate into freshwater to breed, and after spawning they:

- A. Return immediately to the sea
- B. Die soon after spawning ✓**
- C. Continue to live in freshwater for several more years
- D. Undergo metamorphosis into a sessile adult form

Solution: Adult lampreys migrate to freshwater to spawn and die shortly afterwards; their larvae (ammocoetes) later migrate to the sea after metamorphosis.

Question 41

HARD

Assertion (A): Cyclostomata is treated as a class distinct from Pisces even though both possess a notochord.

Reason (R): Cyclostomes lack jaws and paired appendages/fins, features that are present in true fishes.

- A. Both A and R are true and R is the correct explanation of A. ✓**
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false.
- D. A is false but R is true.

Solution: Both statements are correct; the absence of jaws and paired fins in cyclostomes is precisely why they are placed in a separate class from true (jawed) fishes.

Superclass Pisces

Question 42

HARD

Which of the following features is characteristic of class Chondrichthyes but ABSENT in most Osteichthyes?

- A. An operculum covering the gill slits
- B. An air bladder for buoyancy
- C. Placoid scales with a cartilaginous endoskeleton ✓**
- D. A homocercal tail fin

Solution: Chondrichthyans (cartilaginous fishes) have placoid scales and a cartilage-only skeleton, and typically lack both an operculum and an air bladder, unlike bony fishes (Osteichthyes).

Question 43

HARD

Assertion (A): Most cartilaginous fishes are viviparous or ovoviviparous.

Reason (R): They lack an air bladder and must swim continuously to avoid sinking.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is NOT the correct explanation of A. ✓

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are individually true, but the lack of an air bladder (which relates to buoyancy) does not causally explain the mode of reproduction (viviparity) — so R does not correctly explain A.

Question 44

HARD

Select the correct combination of EXCLUSIVELY marine cartilaginous fishes.

A. Scoliodon, Pristis, Torpedo ✓

B. Exocoetus, Hippocampus, Labeo

C. Catla, Clarias, Anabas

D. Betta, Pterophyllum, Gambusia

Solution: Scoliodon (dogfish), Pristis (saw fish) and Torpedo (electric ray) are all cartilaginous, exclusively marine fishes; the other options list bony, largely freshwater fishes.

Question 45

HARD

Which of the following bony fishes is correctly matched with its common name?

A. Hippocampus – Flying fish

B. Exocoetus – Sea horse

C. Pterophyllum – Angel fish ✓

D. Anguilla – Devil ray

Solution: Pterophyllum is commonly known as angel fish. Hippocampus is the sea horse, Exocoetus is the flying fish, and Anguilla is the freshwater eel (not devil ray).

Question 46

HARD

Regarding the caudal (tail) fin, which statement correctly distinguishes the two classes of Pisces?

A. Chondrichthyes typically show a heterocercal tail; most Osteichthyes show a homocercal tail ✓

B. Chondrichthyes typically show a homocercal tail; Osteichthyes show a heterocercal tail

C. Both classes exclusively possess a heterocercal tail

D. Tail fin shape shows no correlation with fish class

Solution: Cartilaginous fishes (Chondrichthyes) generally have an asymmetrical heterocercal tail, while most bony fishes (Osteichthyes) have a symmetrical homocercal tail.

Class Amphibia

Question 47

HARD

Which of the following statements about class Amphibia is INCORRECT?

A. The skin is moist and generally devoid of scales.

B. The heart is three-chambered in most amphibians.

C. All amphibians possess a bony exoskeleton of overlapping scales similar to reptiles. ✓

D. Respiration may occur through skin, buccal cavity, gills or lungs depending on the species/stage.

Solution: Unlike reptiles, amphibians characteristically lack a scaly, cornified exoskeleton — their skin is moist and, in most forms, scale-free, permitting cutaneous respiration.

Question 48

HARD

Assertion (A): Amphibians such as frogs are said to lead a 'dual life' (aquatic and terrestrial).

Reason (R): Their eggs are laid in water and larvae respire through gills, while adults are largely terrestrial and respire through lungs.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct, and the aquatic larval stage followed by a largely terrestrial adult stage is exactly what gives amphibians their characteristic 'dual life'.

Question 49

HARD

Ichthyophis, a limbless, burrowing amphibian, belongs to which order?

A. Anura

B. Urodela (Caudata)

C. Apoda (Gymnophiona) ✓

D. Salientia

Solution: Ichthyophis is a caecilian, belonging to order Apoda (Gymnophiona) — limbless amphibians adapted for a burrowing lifestyle.

Class Reptilia

Question 50

HARD

Which of the following reptiles possesses a nearly four-chambered heart, unlike most other members of the class?

A. Chelone

B. Calotes

C. Crocodilus ✓

D. Hemidactylus

Solution: Crocodiles (Crocodilus) are unique among reptiles in having a nearly complete four-chambered heart, whereas most other reptiles have a three-chambered heart.

Question 51

HARD

Assertion (A): Reptiles are classified as poikilothermic ('cold-blooded') animals.

Reason (R): They lack a physiological mechanism to regulate internal body temperature independent of the environment.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the inability to physiologically regulate body temperature is precisely why reptiles are termed poikilothermic.

Question 52

HARD

The dry, cornified skin covered with epidermal scales in reptiles primarily serves to:

A. Facilitate cutaneous respiration

B. Prevent water loss from the body ✓

C. Aid in gaseous exchange

D. Increase buoyancy in water

Solution: The dry, keratinised, scaly skin of reptiles is a key terrestrial adaptation that minimises water loss through the body surface.

Class Aves

Question 53

HARD

Which of the following statements about class Aves is INCORRECT?

A. The body is streamlined, with forelimbs modified into wings.

B. The heart is four-chambered.

C. All modern birds possess teeth embedded in their jaws for feeding. ✓

D. Bones are pneumatic, reducing body weight for flight.

Solution: Modern birds are toothless; the jaws are modified into a beak. Teeth were present in ancestral/fossil birds like Archaeopteryx but not in extant Aves.

Question 54

HARD

Assertion (A): Birds are endothermic ('warm-blooded') animals.

Reason (R): They can maintain a constant internal body temperature largely independent of the external environment.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; maintaining a constant internal body temperature independent of surroundings is exactly what defines endothermy in birds.

Question 55

HARD

Struthio camelus (ostrich), a flightless bird, belongs to which group?

A. Carinatae (flying birds with a keeled sternum)

B. Ratitae (flightless birds with a flat, keel-less sternum) ✓

C. An exclusively aquatic group of Neornithes

D. The Archaeopteryx group of fossil birds

Solution: Flightless, running birds like the ostrich lack a keeled sternum and are grouped under Ratitae, distinct from the flying, keel-bearing Carinatae.

Class Mammalia

Question 56

HARD

Which of the following mammals is an oviparous (egg-laying) member of subclass Prototheria?

A. Macropus (kangaroo)

B. Ornithorhynchus (platypus) ✓

C. Pteropus (bat)

D. Balaenoptera (whale)

Solution: Ornithorhynchus (duck-billed platypus), along with Echidna, belongs to Prototheria — the only egg-laying mammals. Kangaroo is a marsupial (Metatheria); bat and whale are placental (Eutheria).

Question 57

HARD

Assertion (A): Marsupials such as the kangaroo give birth to poorly developed young that complete development inside a pouch.

Reason (R): Marsupials lack a well-developed placenta for prolonged intrauterine nourishment of the embryo.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the poorly developed (rudimentary) placenta of marsupials is exactly why the young are born early and must complete development attached to a teat inside the pouch.

Question 58

HARD

The presence of heterodont, thecodont teeth (differentiated types, each socketed in the jawbone) is a diagnostic feature most strongly associated with:

A. Class Aves

B. Class Reptilia (in general)

C. Class Mammalia ✓

D. Class Amphibia

Solution: Mammals characteristically show heterodont dentition (incisors, canines, premolars, molars) with teeth set in individual sockets (thecodont) — a combination not typical of the other classes listed.

Question 59

HARD

Which of the following combinations correctly represents EXCLUSIVELY aquatic mammals?

A. Delphinus, Balaenoptera, Platanista ✓

B. Pteropus, Macropus, Camelus

C. Ornithorhynchus, Hemidactylus, Chelone

D. Loris, Pteropus, Vespertilio

Solution: Delphinus (dolphin), Balaenoptera (whale) and Platanista (river dolphin) are all exclusively aquatic mammals. The other options mix in bats, kangaroos, camels, or non-mammals.

Question 60

HARD

Which of the following statements about class Mammalia is INCORRECT?

A. Mammary glands are present in females and used for nourishing the young.

B. All mammals, without exception, are viviparous. ✓

C. Pinna (external ear) is a characteristic external structure.

D. The heart is four-chambered.

Solution: Prototherian mammals such as the platypus (*Ornithorhynchus*) and echidna (*Tachyglossus*) are oviparous (egg-laying), so it is incorrect to say ALL mammals are viviparous.
